NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1) **MA1513** **Lecture Problem Set 5**

Instructions

- To be discussed during lecture in week 5 (Sep 7 and 8, 2026).
- Note that you do not need to submit the answers to this problem set.
- To prepare for the problem set, students should view the "Week 5 key video" before the lecture.
- Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.

1. It is known that a 3×3 matrix $\boldsymbol{A}$ has two eigenvalues -1 and 2 with eigenspaces $E_{-1} = \text{span}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$ and $E_2 = \text{span}\left\{\begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}2\\1\\0\end{pmatrix}\right\}$.

   Without finding the matrix $\boldsymbol{A}$,

   (a) write down the characteristic polynomial of $\boldsymbol{A}$;

   (b) write down a matrix $\boldsymbol{P}$ such that $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \begin{pmatrix}2 & 0 & 0\\0 & -1 & 0\\0 & 0 & 2\end{pmatrix}$;

   (c) find a triangular matrix $\boldsymbol{Q}$ such that $\boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}$ is a diagonal matrix, and write down this diagonal matrix.

2. For the species of animals with two life stages (juvenile and adult) in Problem set 4 with the same stage matrix $\begin{pmatrix}0 & 1.5\\0.4 & 0.7\end{pmatrix}$, suppose the current populations j_0 and a_0 for juveniles and adults are respectively 15,000 and 10,000.

   (a) Find the two populations j_n and a_n in terms of n after n years.

   (b) Will the populations be stabilized in the long term? Why?

   (c) Will the ratio of juveniles to adults be stabilized in the long term? If yes, what is this ratio?

3. For each of the following SDE,

(a) $\begin{cases} y_1' = y_1 + 2y_2 \\ y_2' = 0.5y_1 + y_2 \end{cases}$

(b) $\begin{cases} y_1' = 2y_1 + 2y_2 + 2y_3 \\ y_2' = \quad - 2y_3 \\ y_3' = \quad - 2y_2 \end{cases}$

(i) find a fundamental set of solutions;

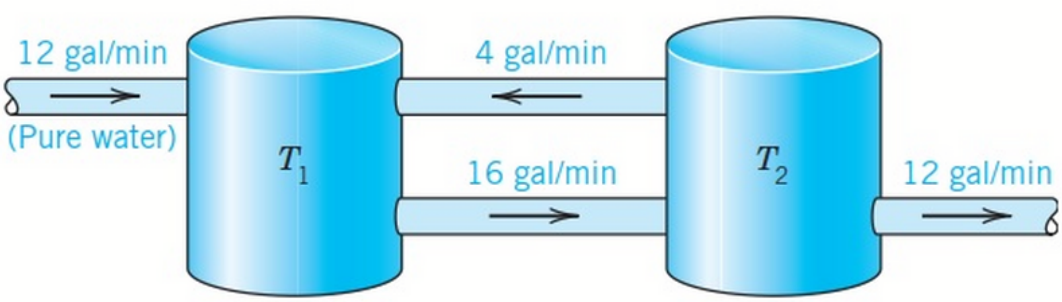
(ii) write down the general solution using the answers in (i);

(iii) find the particular solution of the initial value problem with the initial condition:
(a) $y_1(0) = 1, y_2(0) = 2$ for the 2×2 SDE;
(b) $y_1(0) = 0, y_2(0) = 2, y_3(0) = 0$ for the 3×3 SDE.

4. Consider the two tanks shown in the diagram below. Each tank initially holds 200 gallon of water in which 5 kg of salt has been dissolved. Pure water flows into tank 1 at a rate of 12 gallon/min and the well-stirred mixture flows into tank 2 at a rate of 16 gallon/min. The well stirred mixture in tank 2 is simultaneously pumped back into tank 1 at a rate of 4 gallon/min and out of tank 2 at a rate of 12 gallon/min.

(a) Determine the amount of salt in the two tanks at time t (min).

(b) What will be the amount of salt in each tank in the long run?



Hints and selected answers

1. (a) The eigenvalues are the roots of the characteristic polynomial.
   (b) Find three linearly independent eigenvectors.
   (c) Find another basis for E_2.

2. (a) $j_n = \dfrac{1000}{17}\left(30\left(-\frac{1}{2}\right)^n + 225\left(\frac{6}{5}\right)^n\right)$; $a_n = \dfrac{1000}{17}\left(-10\left(-\frac{1}{2}\right)^n + 180\left(\frac{6}{5}\right)^n\right)$.
   (b) No;
   (c) Ratio 5/4.

3. Find the eigenvalues and eigenvectors for the underlying matrix in each case to form a fundamental set of solutions for the system.
   (a) $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t}$
   Particular solution: $y_1 = -\frac{3}{2} + \frac{5}{2}e^{2t}$, $y_2 = \frac{3}{4} + \frac{5}{4}e^{2t}$

   (b) Like the 2×2 SDE, we can construct solutions of 3×3 SDE by using the eigenvalues and eigenvectors in a similar way. In this case, the fundamental set of solutions consist of three independent solutions.

4. Refer to chapter 4 notes section 4.1 and 4.2.
   Solution: $y_1 = 1.25e^{-0.12t} + 3.75e^{-0.04t}$; $y_2 = -2.5e^{-0.12t} + 7.5e^{-0.04t}$.